



Cadence Driver Verification Plan and Test Report

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Cadence Driver Verification Plan and Test Report

Cadence Design Systems

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Hardware Identifier:015fd689f2e4ddb88dc401f8c26532e

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Chapter 1. Overview

1.1. Document Purpose

The purpose of this document is to define the API testing for the Cadence driver firmware and to present the results.

1.2. Reference Documents

- IEEE 802.1Q - IEEE Standard for Local and metropolitan area networks Virtual Bridged Local Area Networks
- IEE 1588 - Precision clock synchronisation (PTP) v1 - 2002 and v2 - 2008

1.3. Acronyms

API	Application Programming Interface.
Core Driver	Cadence Firmware component that provides IP programming abstraction.
DMA	Direct Memory Access
EDD	Ethernet DMA-MAC Driver
EMAC	Ethernet MAC hardware IP
MAC	Media Access Control
SoC	System on a Chip

Chapter 2. Test Equipment/Environment

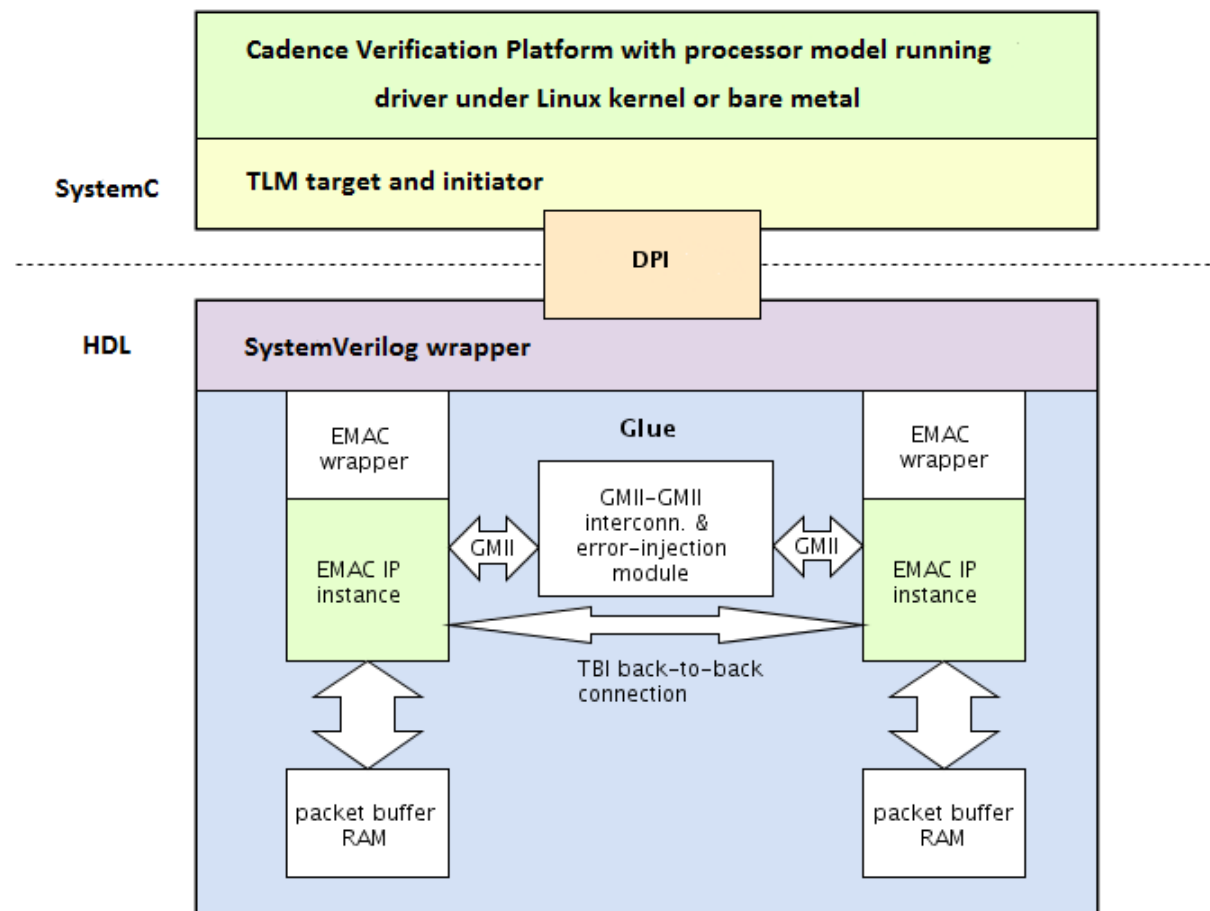
2.1. Overview

All testing is on a hardware simulated platform, running either a bare-metal or Linux build. A sequence of tests is automatically generated as an executable program during the build process, and run from within the Cadence firmware verification platform environment. Results are logged to a file via the Cadence firmware verification platform console output then parsed using a python script and formatted into this report.

2.2. Cadence firmware verification platform environment

The simulated hardware environment utilizes the Cadence firmware verification platform to connect the driver to the actual RTL for your EMAC device. The architecture is as follows:

Figure 2.1. Environment for verification of the EMAC FW driver



2.3. Driver Tests

These tests confirm the operation of the core driver API under a range of different inputs, ensuring that operation complies with that described in the user documentation and that there are no unexpected errors. Some tests may not be appropriate for some APIs, depending on configuration, if so these will be marked as "NOT SUPPORTED". For each test which runs, the result will be displayed as "PASSED" or "FAILED". Tests include:

Test Equipment/Environment

Null pointer tests	API is checked to confirm that all null pointers are detected, and return appropriate error code
Get/set and register access tests	API is checked for all get/set or register access functions, and confirm that these can correctly write and read the value.
Ranges tests	API is checked with valid and invalid parameters, and return code compared with that expected.
Functional tests	A sequence of predefined tests is run, with an expected result. These tests exercise the Cadence IP core, and may require use of additional test IP in the verification platform.

NOTE: Not all configurations will require all types of test.

Chapter 3. Test Results

3.1. Test Summary

FunctionalTest: 66 calls, 48 passed, 18 not supported

nullptr: 274 calls, 274 passed

getset: 88 calls, 59 passed, 29 not supported

ranges: 274 calls, 184 passed, 90 not supported

3.2. Test Details

3.2.1. FunctionalTest

```
highLoadTest PASSED
probeTest PASSED
initTest PASSED
txTest PASSED
txDStatDecodeTest PASSED
callbacksTest PASSED
txTestMul PASSED
txTestFreeDesc PASSED
rxReadBufTest PASSED
rxAddBufTest PASSED
txRxTest PASSED
cbTxRxTest PASSED
cbTxRxTest2 PASSED
rxDStatDecodeTest PASSED
rxFiltTest1 PASSED
rxFiltTest2 PASSED
rxFiltTest3 PASSED
txRxPollTest PASSED
txRxEventTest PASSED
mdioRwTest PASSED
rxLengthTest PASSED
txRxAllQsTest PASSED
tsuTimerTest PASSED
jumboFramesTest PASSED
priorityTxQueuesTest NOT SUPPORTED
Need at least 3 priority queues for this test:
rxScrnTest1 NOT SUPPORTED
Need multiple queues for this test:
rxScrnTest2 NOT SUPPORTED
:
rxScrnTest3 NOT SUPPORTED
:
ptpEventDetectTest PASSED
rxOverrunTest PASSED
pauseDetectTest PASSED
pcsAutonegTest_1G PASSED
loopbackTest NOT SUPPORTED
Not available when in PCS mode:
ptpSingleStepTest PASSED
timeStampTxRxTest PASSED
descTimeStampModeTest PASSED
dmaUpper32AddrTest PASSED
statsApiTest PASSED
largeSendOffloadsTest PASSED
headerDataSplitTest PASSED
```

Test Results

rscTest NOT SUPPORTED
Need multiple queues for this test:
intrptModerationTest PASSED
multiFrameTxRx PASSED
resetRxQTest NOT SUPPORTED
Need multiple queues for this test:
swDisableEnableRxQsTest NOT SUPPORTED
Need multiple queues for this test:
EnstTest NOT SUPPORTED
Need at least 3 priority queues for this test:
swDisableEnableTxQsTest NOT SUPPORTED
Need multiple queues for this test:
reportBadFcsTest PASSED
pcsAutonegTest_2.5G PASSED
miiOnRgmiiTest PASSED
frerRegAccessTest NOT SUPPORTED
Frame Elimination is not available in this configuration:
frerEliminationTest NOT SUPPORTED
Frame Elimination is not available in this configuration:
emacTxRxTest NOT SUPPORTED
Interspersing Express Traffic is not available or not enabled:
emacTxPriorityTest NOT SUPPORTED
Interspersing Express Traffic is not available or not enabled:
txTestPartFrameFreeDesc PASSED
rxScrn1DropTest NOT SUPPORTED
:
rxScrn2DropTest NOT SUPPORTED
:
queueReceiveFlushTest PASSED
rxScrn2RateLimitTest NOT SUPPORTED
Need multiple queues for this test:
intrptModerationThresholdTest PASSED
autoTransPauseTest PASSED
lockupRegAccessTest PASSED
timeSchedTest PASSED
vlanMatchByIdTest NOT SUPPORTED
T2 compare offset position S-TAG or C-TAG is not supported
getLinkFaultIndicationTest PASSED
asfSimpleTest PASSED

3.2.2. nullptr

probe PASSED
init PASSED
destroy PASSED
start PASSED
stop PASSED
isr PASSED
setEventEnable PASSED
getEventEnable PASSED
setIntrptModerate PASSED
getIntrptModerate PASSED
setIntrptModerateThreshold PASSED
getIntrptModerateThreshold PASSED
setIfSpeed PASSED
getIfSpeed PASSED
setJumboFramesRx PASSED
getJumboFramesRx PASSED
setJumboFrameRxMaxLen PASSED
getJumboFrameRxMaxLen PASSED
setUniDirEnable PASSED
getUniDirEnable PASSED
setTxChecksumOffload PASSED

Test Results

getTxChecksumOffload PASSED
setRxBufOffset PASSED
getRxBufOffset PASSED
set1536ByteFramesRx PASSED
get1536ByteFramesRx PASSED
setRxChecksumOffload PASSED
getRxChecksumOffload PASSED
setFcsRemove PASSED
getFcsRemove PASSED
setTxPartialStFwd PASSED
getTxPartialStFwd PASSED
setRxPartialStFwd PASSED
getRxPartialStFwd PASSED
setRxDmaDataAddrMask PASSED
getRxDmaDataAddrMask PASSED
setRxBadPreamble PASSED
getRxBadPreamble PASSED
setFullDuplex PASSED
getFullDuplex PASSED
setIgnoreFcsRx PASSED
getIgnoreFcsRx PASSED
setRxHalfDuplexInTx PASSED
getRxHalfDuplexInTx PASSED
getIfCapabilities PASSED
setLoopback PASSED
getLoopback PASSED
calcMaxTxFrameSize PASSED
queueTxBuf PASSED
qTxBuf PASSED
deQTxBuf PASSED
txDescFree PASSED
freeTxDesc PASSED
getTxDescStat PASSED
getTxDescSize PASSED
resetTxQ PASSED
startTx PASSED
stopTx PASSED
abortTx PASSED
transmitting PASSED
enableTx PASSED
getTxEnabled PASSED
getTxStatus PASSED
clearTxStatus PASSED
enableCbs PASSED
disableCbs PASSED
getCbsQSetting PASSED
setIpgStretch PASSED
getIpgStretch PASSED
calcMaxRxFrameSize PASSED
addRxBuf PASSED
numRxBufs PASSED
numRxUsed PASSED
readRxBuf PASSED
getRxDescStat PASSED
getRxDescSize PASSED
rxEnabled PASSED
enableRx PASSED
disableRx PASSED
removeRxBuf PASSED
resetRxQ PASSED
getRxStatus PASSED
clearRxStatus PASSED
setHdrDataSplit PASSED
getHdrDataSplit PASSED

Test Results

setRscEnable PASSED
getRscEnable PASSED
setRscClearMask PASSED
setSpecificAddr PASSED
getSpecificAddr PASSED
setSpecificAddr1Mask PASSED
getSpecificAddr1Mask PASSED
disableSpecAddr PASSED
setTypeIdMatch PASSED
getTypeIdMatch PASSED
setUnicastEnable PASSED
getUnicastEnable PASSED
setMulticastEnable PASSED
getMulticastEnable PASSED
setNoBroadcast PASSED
getNoBroadcast PASSED
setVlanOnly PASSED
getVlanOnly PASSED
setStackedVlanReg PASSED
getStackedVlanReg PASSED
setCopyAllFrames PASSED
getCopyAllFrames PASSED
setHashAddr PASSED
getHashAddr PASSED
setLenErrDiscard PASSED
getLenErrDiscard PASSED
getNumScreenRegs PASSED
setType1ScreenReg PASSED
getType1ScreenReg PASSED
setType2ScreenReg PASSED
getType2ScreenReg PASSED
setType2EthertypeReg PASSED
getType2EthertypeReg PASSED
setType2CompareReg PASSED
getType2CompareReg PASSED
setPauseEnable PASSED
getPauseEnable PASSED
txPauseFrame PASSED
txZeroQPause PASSED
getRxPauseQuantum PASSED
setTxPauseQuantum PASSED
getTxPauseQuantum PASSED
setCopyPauseDisable PASSED
getCopyPauseDisable PASSED
setPfcPriorityBasedPauseRx PASSED
getPfcPriorityBasedPauseRx PASSED
txPfcPriorityBasedPause PASSED
setTxPfcPauseFrameFields PASSED
getTxPfcPauseFrameFields PASSED
setEnableMultiPfcPauseQuantum PASSED
getEnableMultiPfcPauseQuantum PASSED
setUnicastPtpDetect PASSED
getUnicastPtpDetect PASSED
setPtpRxUnicastIpAddr PASSED
getPtpRxUnicastIpAddr PASSED
setPtpTxUnicastIpAddr PASSED
getPtpTxUnicastIpAddr PASSED
set1588Timer PASSED
get1588Timer PASSED
adjust1588Timer PASSED
set1588TimerInc PASSED
get1588TimerInc PASSED
setTsuTimerCompVal PASSED
getTsuTimerCompVal PASSED

Test Results

getPtpFrameTxTime PASSED
getPtpFrameRxTime PASSED
getPtpPeerFrameTxTime PASSED
getPtpPeerFrameRxTime PASSED
get1588SyncStrobeTime PASSED
setExtTsuPortEnable PASSED
getExtTsuPortEnable PASSED
set1588OneStepTxSyncEnable PASSED
get1588OneStepTxSyncEnable PASSED
setDescTimeStampMode PASSED
getDescTimeStampMode PASSED
setStoreRxTimeStamp PASSED
getStoreRxTimeStamp PASSED
resetPcs PASSED
getPcsReady PASSED
startAutoNegotiation PASSED
setAutoNegEnable PASSED
getAutoNegEnable PASSED
getLinkStatus PASSED
getAnRemoteFault PASSED
getAnComplete PASSED
setAnAdvPage PASSED
getAnAdvPage PASSED
getLpAbilityPage PASSED
getPageRx PASSED
setNextPageTx PASSED
getNextPageTx PASSED
getLpNextPage PASSED
getPhyId PASSED
setMdioEnable PASSED
getMdioEnable PASSED
phyStartMdioWrite PASSED
phyStartMdioRead PASSED
getMdioReadData PASSED
getMdioIdle PASSED
readStats PASSED
clearStats PASSED
takeSnapshot PASSED
setReadSnapshot PASSED
getReadSnapshot PASSED
setWakeOnLanReg PASSED
getWakeOnLanReg PASSED
setLpiTxEnable PASSED
getLpiTxEnable PASSED
getLpiStats PASSED
getDesignConfig PASSED
setWriteStatsEnable PASSED
getWriteStatsEnable PASSED
incStatsRegs PASSED
setRxBackPressure PASSED
getRxBackPressure PASSED
setCollisionTest PASSED
getCollisionTest PASSED
setRetryTest PASSED
getRetryTest PASSED
writeUserOutputs PASSED
readUserOutputs PASSED
setUserOutPin PASSED
readUserInputs PASSED
getUserInPin PASSED
getMdioInState PASSED
readReg PASSED
writeReg PASSED
setTxQueueNum PASSED

Test Results

setRxQueueNum PASSED
setTxQueueSegAlloc PASSED
setTxQueueScheduler PASSED
getTxQueueScheduler PASSED
setDwrrWeighting PASSED
setEtsBandAlloc PASSED
getDwrrWeighting PASSED
getEtsBandAlloc PASSED
setEnstTimeConfig PASSED
getEnstTimeConfig PASSED
setEnstEnable PASSED
getEnstEnable PASSED
getEnstSupported PASSED
setReportingBadFCS PASSED
getReportingBadFCS PASSED
setPtpSingleStep PASSED
getPtpSingleStep PASSED
setMiiOnRgmii PASSED
getMiiOnRgmii PASSED
setCbsIdleSlope PASSED
getCbsIdleSlope PASSED
getLinkFaultIndication PASSED
setFrameEliminationTagSize PASSED
getFrameEliminationTagSize PASSED
setFrameEliminationEnable PASSED
getFrameEliminationEnable PASSED
setFrameEliminationSeqRecRstTmrEnable PASSED
getFrameEliminationSeqRecRstTmrEnable PASSED
setFrameEliminationConfig PASSED
getFrameEliminationConfig PASSED
setFrameEliminationTagConfig PASSED
getFrameEliminationTagConfig PASSED
setFrameEliminationTimeoutConfig PASSED
getFrameEliminationStats PASSED
getEmac PASSED
getPmac PASSED
getMacType PASSED
setPreemptionConfig PASSED
getPreemptionConfig PASSED
preemptionRestartVerification PASSED
setMmslEventEnable PASSED
getMmslEventEnable PASSED
readMmslStats PASSED
readMmslStatus PASSED
setLockupConfig PASSED
getLockupConfig PASSED
setType1ScreenRegDropEnable PASSED
getType1ScreenRegDropEnable PASSED
setType2ScreenRegDropEnable PASSED
getType2ScreenRegDropEnable PASSED
setRxQFlushConfig PASSED
getRxQFlushConfig PASSED
getRxDmaFlushedPacketsCount PASSED
setType2ScreenerRateLimit PASSED
getType2ScreenerRateLimit PASSED
getRxType2RateLimitTriggered PASSED
setRxWatermark PASSED
getRxWatermark PASSED
setAxiQosConfig PASSED
getAxiQosConfig PASSED
getAsfInfo PASSED

3.2.3. getset

```

EventEnable NOT SUPPORTED
IntrptModerate PASSED
IntrptModerateThreshold PASSED
IfSpeed PASSED
JumboFramesRx PASSED
JumboFrameRxMaxLen PASSED
UniDirEnable PASSED
TxChecksumOffload PASSED
RxBufOffset PASSED
1536ByteFramesRx PASSED
RxChecksumOffload PASSED
FcsRemove PASSED
TxPartialStFwd PASSED
RxPartialStFwd PASSED
RxDmaDataAddrMask PASSED
RxBadPreamble PASSED
FullDuplex PASSED
IgnoreFcsRx PASSED
RxHalfDuplexInTx PASSED
Loopback NOT SUPPORTED
IpgStretch PASSED
HdrDataSplit PASSED
RscEnable NOT SUPPORTED
SpecificAddr PASSED
SpecificAddr1Mask NOT SUPPORTED
TypeIdMatch PASSED
UnicastEnable PASSED
MulticastEnable PASSED
NoBroadcast PASSED
VlanOnly PASSED
StackedVlanReg PASSED
CopyAllFrames PASSED
HashAddr PASSED
LenErrDiscard PASSED
Type1ScreenReg PASSED
Type2ScreenReg PASSED
Type2EthertypeReg NOT SUPPORTED
Type2CompareReg NOT SUPPORTED
PauseEnable PASSED
TxPauseQuantum NOT SUPPORTED
CopyPauseDisable PASSED
PfcPriorityBasedPauseRx PASSED
TxPfcPauseFrameFields PASSED
EnableMultiPfcPauseQuantum NOT SUPPORTED
UnicastPtpDetect PASSED
PtpRxUnicastIpAddr PASSED
PtpTxUnicastIpAddr PASSED
Not supported by mechanical tests, covered in functional testing
1588Timer NOT SUPPORTED
1588TimerInc NOT SUPPORTED
TsuTimerCompVal NOT SUPPORTED
External TSU not available
ExtTsuPortEnable NOT SUPPORTED
1588OneStepTxSyncEnable NOT SUPPORTED
DescTimeStampMode NOT SUPPORTED
StoreRxTimeStamp NOT SUPPORTED
AutoNegEnable NOT SUPPORTED
AnAdvPage NOT SUPPORTED
NextPageTx NOT SUPPORTED
MdioEnable NOT SUPPORTED
ReadSnapshot PASSED

```

Test Results

WakeOnLanReg PASSED
LpiTxEnable PASSED
WriteStatsEnable PASSED
RxBackPressure PASSED
CollisionTest NOT SUPPORTED
RetryTest PASSED
TxQueueScheduler PASSED
DwrrWeighting PASSED
EtsBandAlloc PASSED
EnstTimeConfig PASSED
EnstEnable PASSED
ReportingBadFCS PASSED
PtpSingleStep PASSED
MiiOnRgmii PASSED
CbsIdleSlope PASSED
FrameEliminationTagSize NOT SUPPORTED
FrameEliminationEnable NOT SUPPORTED
FrameEliminationSeqRecRstTmrEnable NOT SUPPORTED
FrameEliminationConfig NOT SUPPORTED
FrameEliminationTagConfig NOT SUPPORTED
PreemptionConfig NOT SUPPORTED
MmslEventEnable NOT SUPPORTED
LockupConfig PASSED
Type1ScreenRegDropEnable NOT SUPPORTED
Type2ScreenRegDropEnable NOT SUPPORTED
RxQFlushConfig PASSED
Type2ScreenerRateLimit PASSED
RxWatermark PASSED
AxiQosConfig PASSED

3.2.4. ranges

Not in mechanical tests, covered by functional testing
probe NOT SUPPORTED
Not in mechanical tests, covered by functional testing
init NOT SUPPORTED
destroy PASSED
start PASSED
stop PASSED
isr PASSED
setEventEnable PASSED
getEventEnable PASSED
setIntrptModerate PASSED
getIntrptModerate PASSED
setIntrptModerateThreshold PASSED
getIntrptModerateThreshold PASSED
setIfSpeed PASSED
getIfSpeed PASSED
setJumboFramesRx PASSED
getJumboFramesRx PASSED
setJumboFrameRxMaxLen PASSED
getJumboFrameRxMaxLen PASSED
setUniDirEnable PASSED
getUniDirEnable PASSED
setTxChecksumOffload PASSED
getTxChecksumOffload PASSED
setRxBufOffset PASSED
getRxBufOffset PASSED
set1536ByteFramesRx PASSED
get1536ByteFramesRx PASSED
setRxChecksumOffload PASSED
getRxChecksumOffload PASSED
setFcsRemove PASSED

Test Results

getFcsRemove PASSED
setTxPartialStFwd PASSED
getTxPartialStFwd PASSED
setRxPartialStFwd PASSED
getRxPartialStFwd PASSED
setRxDmaDataAddrMask PASSED
getRxDmaDataAddrMask PASSED
setRxBadPreamble PASSED
getRxBadPreamble PASSED
setFullDuplex PASSED
getFullDuplex PASSED
setIgnoreFcsRx PASSED
getIgnoreFcsRx PASSED
setRxHalfDuplexInTx PASSED
getRxHalfDuplexInTx PASSED
getIfCapabilities PASSED
setLoopback NOT SUPPORTED
getLoopback PASSED
calcMaxTxFrameSize PASSED
Return depends on dynamic variable nFree
queueTxBuf NOT SUPPORTED
Return depends on dynamic variable nFree
qTxBuf NOT SUPPORTED
Return depends on state of descriptor queue
deqTxBuf NOT SUPPORTED
txDescFree PASSED
freeTxDesc PASSED
getTxDescStat PASSED
getTxDescSize PASSED
resetTxQ PASSED
startTx PASSED
stopTx PASSED
abortTx PASSED
transmitting PASSED
enableTx PASSED
getTxEnabled PASSED
getTxStatus PASSED
clearTxStatus PASSED
enableCbs PASSED
disableCbs PASSED
getCbsQSetting PASSED
setIpgStretch PASSED
getIpgStretch PASSED
calcMaxRxFrameSize PASSED
Not supported
addRxBuf NOT SUPPORTED
numRxBufs PASSED
numRxUsed PASSED
Not supported, covered by functional testing
readRxBuf NOT SUPPORTED
getRxDescStat PASSED
getRxDescSize PASSED
rxEnabled PASSED
enableRx PASSED
disableRx PASSED
Not supported
removeRxBuf NOT SUPPORTED
resetRxQ PASSED
getRxStatus PASSED
clearRxStatus PASSED
setHdrDataSplit PASSED
getHdrDataSplit PASSED
setRscEnable NOT SUPPORTED
getRscEnable PASSED

Test Results

setRscClearMask NOT SUPPORTED
setSpecificAddr PASSED
getSpecificAddr NOT SUPPORTED
setSpecificAddr1Mask NOT SUPPORTED
getSpecificAddr1Mask NOT SUPPORTED
disableSpecAddr NOT SUPPORTED
setTypeIdMatch PASSED
getTypeIdMatch PASSED
setUnicastEnable PASSED
getUnicastEnable PASSED
setMulticastEnable PASSED
getMulticastEnable PASSED
setNoBroadcast PASSED
getNoBroadcast PASSED
setVlanOnly PASSED
getVlanOnly PASSED
setStackedVlanReg PASSED
getStackedVlanReg PASSED
setCopyAllFrames PASSED
getCopyAllFrames PASSED
setHashAddr PASSED
getHashAddr PASSED
setLenErrDiscard PASSED
getLenErrDiscard PASSED
getNumScreenRegs PASSED
setType1ScreenReg NOT SUPPORTED
getType1ScreenReg NOT SUPPORTED
setType2ScreenReg NOT SUPPORTED
getType2ScreenReg NOT SUPPORTED
setType2EthertypeReg NOT SUPPORTED
getType2EthertypeReg NOT SUPPORTED
setType2CompareReg NOT SUPPORTED
getType2CompareReg NOT SUPPORTED
setPauseEnable PASSED
getPauseEnable PASSED
txPauseFrame PASSED
txZeroQPause PASSED
getRxPauseQuantum PASSED
setTxPauseQuantum NOT SUPPORTED
getTxPauseQuantum NOT SUPPORTED
setCopyPauseDisable PASSED
getCopyPauseDisable PASSED
setPfcPriorityBasedPauseRx PASSED
getPfcPriorityBasedPauseRx PASSED
Not supported
txPfcPriorityBasedPause NOT SUPPORTED
setTxPfcPauseFrameFields PASSED
getTxPfcPauseFrameFields PASSED
setEnableMultiPfcPauseQuantum NOT SUPPORTED
getEnableMultiPfcPauseQuantum PASSED
setUnicastPtpDetect PASSED
getUnicastPtpDetect PASSED
setPtpRxUnicastIpAddr PASSED
getPtpRxUnicastIpAddr PASSED
setPtpTxUnicastIpAddr PASSED
getPtpTxUnicastIpAddr PASSED
set1588Timer NOT SUPPORTED
Not supported by mechanical tests, covered in functional testing
get1588Timer NOT SUPPORTED
adjust1588Timer NOT SUPPORTED
set1588TimerInc NOT SUPPORTED
get1588TimerInc NOT SUPPORTED
setTsuTimerCompVal NOT SUPPORTED
getTsuTimerCompVal NOT SUPPORTED

Test Results

getPtpFrameTxTime NOT SUPPORTED
getPtpFrameRxTime NOT SUPPORTED
getPtpPeerFrameTxTime NOT SUPPORTED
getPtpPeerFrameRxTime NOT SUPPORTED
get1588SyncStrobeTime NOT SUPPORTED
setExtTsuPortEnable NOT SUPPORTED
External TSU not available
getExtTsuPortEnable NOT SUPPORTED
set1588OneStepTxSyncEnable NOT SUPPORTED
get1588OneStepTxSyncEnable PASSED
setDescTimeStampMode PASSED
getDescTimeStampMode NOT SUPPORTED
setStoreRxTimeStamp NOT SUPPORTED
getStoreRxTimeStamp NOT SUPPORTED
resetPcs NOT SUPPORTED
getPcsReady NOT SUPPORTED
startAutoNegotiation NOT SUPPORTED
setAutoNegEnable NOT SUPPORTED
getAutoNegEnable NOT SUPPORTED
getLinkStatus PASSED
getAnRemoteFault PASSED
getAnComplete NOT SUPPORTED
setAnAdvPage NOT SUPPORTED
getAnAdvPage NOT SUPPORTED
getLpAbilityPage NOT SUPPORTED
getPageRx PASSED
setNextPageTx NOT SUPPORTED
getNextPageTx PASSED
getLpNextPage PASSED
getPhyId NOT SUPPORTED
setMdioEnable PASSED
getMdioEnable PASSED
phyStartMdioWrite PASSED
phyStartMdioRead PASSED
getMdioReadData PASSED
Not supported
getMdioIdle NOT SUPPORTED
readStats PASSED
clearStats PASSED
takeSnapshot PASSED
setReadSnapshot PASSED
getReadSnapshot PASSED
setWakeOnLanReg PASSED
getWakeOnLanReg PASSED
setLpiTxEnable PASSED
getLpiTxEnable PASSED
getLpiStats PASSED
getDesignConfig PASSED
setWriteStatsEnable PASSED
getWriteStatsEnable PASSED
incStatsRegs PASSED
setRxBackPressure PASSED
getRxBackPressure PASSED
setCollisionTest NOT SUPPORTED
getCollisionTest NOT SUPPORTED
setRetryTest PASSED
getRetryTest PASSED
writeUserOutputs NOT SUPPORTED
readUserOutputs NOT SUPPORTED
setUserOutPin NOT SUPPORTED
readUserInputs NOT SUPPORTED
getUserInPin NOT SUPPORTED
getMdioInState PASSED
Mechanical tests not supported - diagnostic only

Test Results

readReg NOT SUPPORTED
Mechanical tests not supported - diagnostic only
writeReg NOT SUPPORTED
setTxQueueNum PASSED
setRxQueueNum PASSED
Not in mechanical tests, covered by functional testing
setTxQueueSegAlloc NOT SUPPORTED
setTxQueueScheduler PASSED
getTxQueueScheduler PASSED
setDwrrWeighting PASSED
setEtsBandAlloc PASSED
getDwrrWeighting PASSED
getEtsBandAlloc PASSED
setEnstTimeConfig PASSED
getEnstTimeConfig PASSED
setEnstEnable PASSED
getEnstEnable PASSED
getEnstSupported PASSED
setReportingBadFCS PASSED
getReportingBadFCS PASSED
setPtpSingleStep PASSED
getPtpSingleStep PASSED
setMiiOnRgmii PASSED
getMiiOnRgmii PASSED
setCbsIdleSlope PASSED
getCbsIdleSlope PASSED
getLinkFaultIndication PASSED
setFrameEliminationTagSize PASSED
getFrameEliminationTagSize NOT SUPPORTED
setFrameEliminationEnable NOT SUPPORTED
getFrameEliminationEnable NOT SUPPORTED
setFrameEliminationSeqRecRstTmrEnable NOT SUPPORTED
getFrameEliminationSeqRecRstTmrEnable NOT SUPPORTED
setFrameEliminationConfig NOT SUPPORTED
getFrameEliminationConfig NOT SUPPORTED
setFrameEliminationTagConfig NOT SUPPORTED
getFrameEliminationTagConfig NOT SUPPORTED
setFrameEliminationTimeoutConfig NOT SUPPORTED
getFrameEliminationStats PASSED
getEmac NOT SUPPORTED
getPmac NOT SUPPORTED
getMacType PASSED
setPreemptionConfig NOT SUPPORTED
getPreemptionConfig NOT SUPPORTED
preemptionRestartVerification NOT SUPPORTED
setMmslEventEnable NOT SUPPORTED
getMmslEventEnable NOT SUPPORTED
readMmslStats NOT SUPPORTED
readMmslStatus NOT SUPPORTED
setLockupConfig PASSED
getLockupConfig PASSED
setType1ScreenRegDropEnable NOT SUPPORTED
getType1ScreenRegDropEnable NOT SUPPORTED
setType2ScreenRegDropEnable NOT SUPPORTED
getType2ScreenRegDropEnable NOT SUPPORTED
setRxQFlushConfig PASSED
getRxQFlushConfig PASSED
getRxDmaFlushedPacketsCount PASSED
setType2ScreenerRateLimit PASSED
getType2ScreenerRateLimit PASSED
getRxType2RateLimitTriggered PASSED
setRxWatermark PASSED
getRxWatermark PASSED
setAxiQosConfig PASSED

Test Results

getAxiQosConfig PASSED
getAsfInfo PASSED

Chapter 4. Parasoftware DTP Engine for C/C++ Analysis

Summary:

Parasoftware DTP Engine for C/C++ Analysis: 53 rules detected 0 violations.

Checked rules:

- All exit paths from a function with non-void return type shall have an explicit return statement with an expression
- All member variables should be initialized in constructor
- Always catch exceptions
- Always provide empty brackets ([]) for delete when deallocating arrays
- Avoid accessing arrays out of bounds
- Avoid buffer overflow due to defining incorrect format limits
- Avoid calling virtual functions from constructors
- Avoid conditions that always evaluate to the same value
- Avoid division by zero
- Avoid double locking
- Avoid function duplication
- Avoid infinite loops
- Avoid null pointer dereferencing
- Avoid overflow due to reading a not zero terminated string
- Avoid overflow when reading from a buffer
- Avoid overflow when writing to a buffer
- Avoid race conditions when using fork and file descriptors
- Avoid race conditions while checking for the existence of a symbolic link
- Avoid use before initialization
- Boolean condition always evaluates to the same value due to enumeration with only zero or only non-zero constants
- Class cannot inherit other class more than once unless it is virtual inheritance
- Declare a copy assignment operator for classes with dynamically allocated memory
- Declare a copy constructor for classes with dynamically allocated memory
- Do not abandon unreleased locks
- Do not acquire locks in different order
- Do not call delete on non-pointers
- Do not cast from or to incomplete class at the point of casting
- Do not check for null after dereferencing
- Do not delete objects with incomplete class at the point of deletion
- Do not free resources using invalid pointers
- Do not invoke malloc/realloc for objects having constructors
- Do not pass negative values to functions expecting non-negative arguments
- Do not throw from within destructor
- Do not use blocking functions while holding a lock
- Do not use 'delete' on pointers to a void type
- Do not use resources that have been freed
- Do not use string literals as operands of equality or relational operators
- Ensure resources are freed
- Floating-point expressions shall not be tested for equality or inequality
- If a class has virtual functions it shall have a virtual destructor
- Local variables should not use the same names as member variables
- McCabe Cyclomatic Complexity
- Nested Blocks Depth
- Never provide brackets ([]) for delete when deallocating non-arrays
- Pass objects by reference instead of by value
- Pointer arithmetic performed on freshly allocated memory
- Pointer should not be compared with NULL using relational operators
<, >, >=, <=
- Properly terminate character strings
- Restore stream format
- Suspicious argument to malloc
- Suspicious setting of stream flags
- The address of an object with automatic storage shall not be returned from a

function
Throw by value, catch by reference